

Research of Post-Quantum Cryptography in China

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Quantum Revolution

**Quantum
Communication**

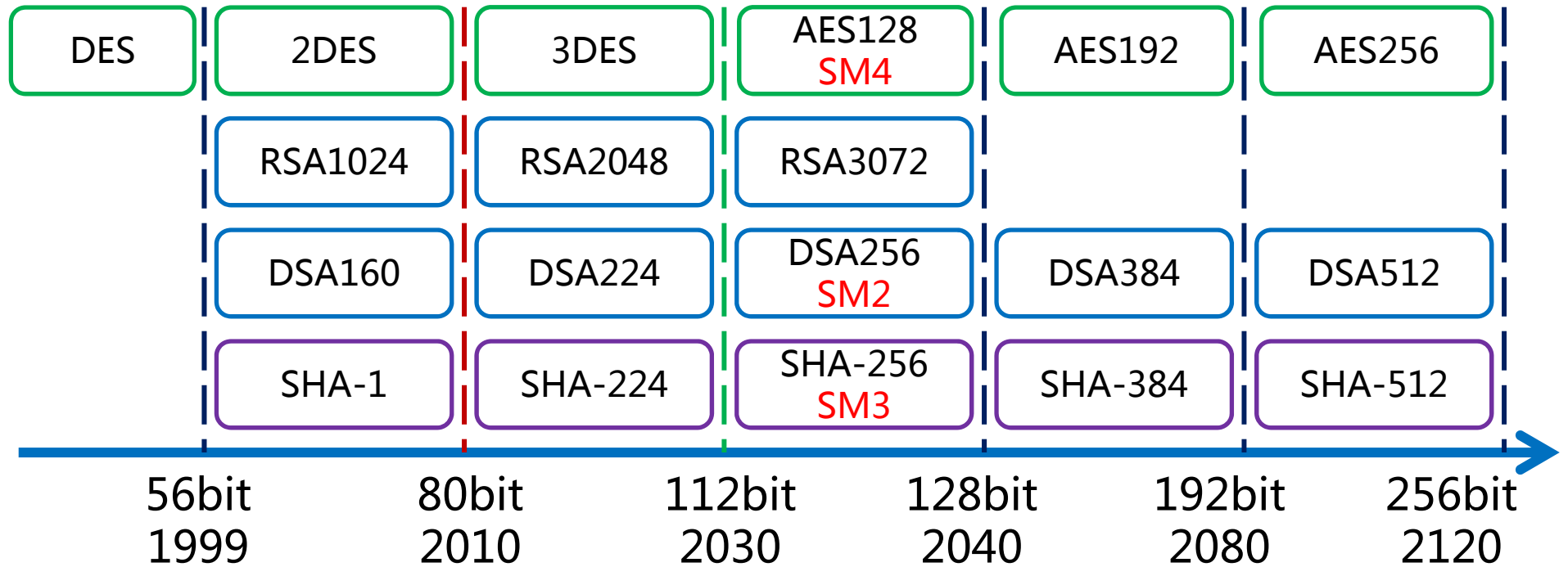
**Quantum
Computation**

Quantum Precision Measurement

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- ☐ Background
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 - ☐ Trends
-

Classical Cryptographic Schemes

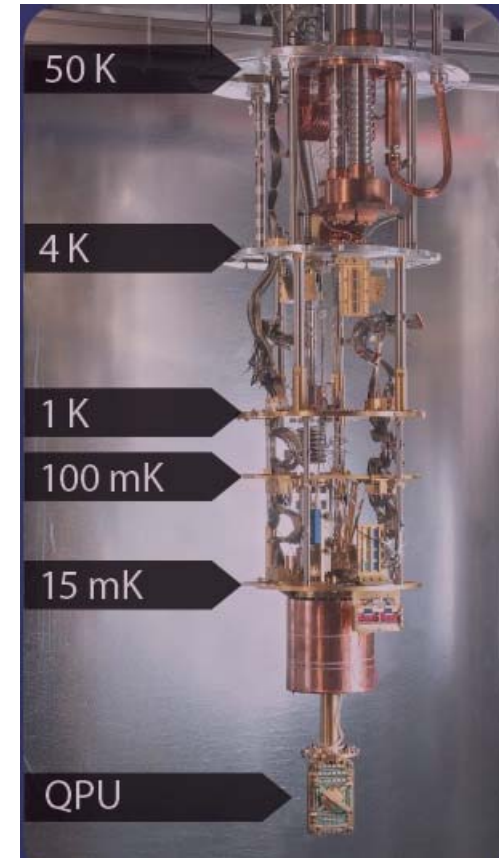


Safe world without quantum computing
Current schemes can be used for 100 years

Quantum Computers



Temporal Defense Systems Inc. (TDS)



Affect of Quantum Computing

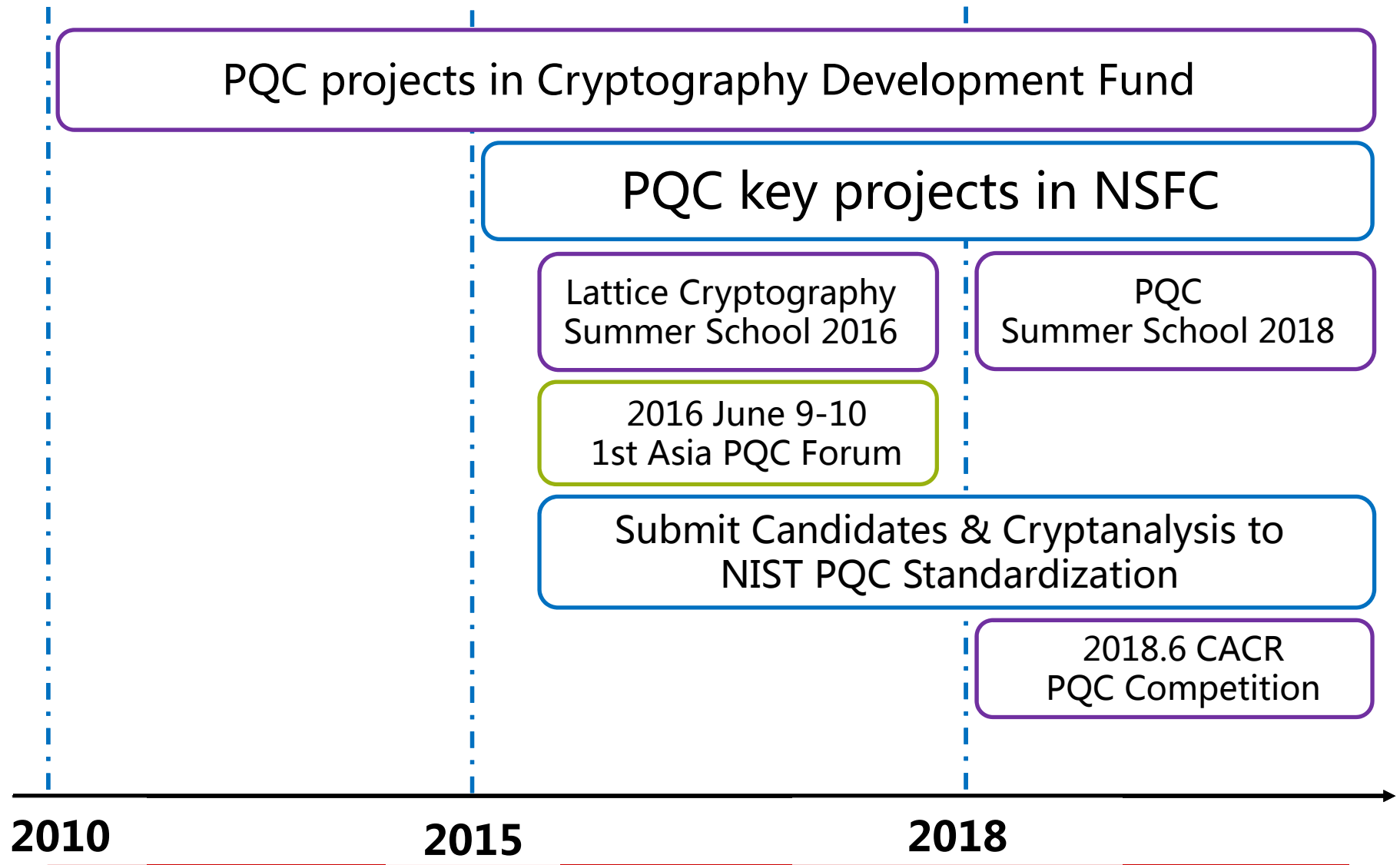
Scheme	Affect
Symmetric Key (SM4,AES)	Security Halved (Grover)
Hash(SM3,SHA-3)	Security Decreased(Grover)
Public Key (RSA,DSA,SM2)	Completely Broken (Shor)
Lattice Cryptography	Quantum Safe (Currently
Multivariant Cryptogrphy	Quantum Safe (Currently
Hash based signature	Quantum Safe (Currently
Code-based cryptography	Quantum Safe (Currently
Isogeny Cryptography	Quantum Safe (Currently

Candidates of NIST PQC

- 37 preliminary submissions (early deadline Sep 2017)
- 82 total submissions received
 - 69 accepted as “complete and proper” (5 since withdrawn)

	Signatures	KEM/Encryption	Overall
Lattice-based	5	21	26
Code-based	2	17	19
Multi-variate	7	2	9
Symmetric/Hash-based	3		3
Other	2	5	7
Total	19	45	64

PQC Events in China



Candidates Submitted to NIST PQC

Algorithms	Inventors
Lepton	Yu yu, Shanghai Jiaotong University, China Zhangjiang, State Key Laboratory of Cryptology, China
KCL	Yunlei Zhao, Zhengzhong jin, Boru Gong, Guangye Sui Fudan University, China
LAC	Xianhui Lu, Yamin Liu, Dingding Jia, Haiyang Xue, Jingnan He DACAS, Chinese Academy of Sciences Zhenfei Zhang, OnBoard Security Inc

1st Candidate Submitted to NIST PQC

Lepton: LPN-based KEMs with Post-Quantum Security

Yu Yu and Jiang Zhang

April 11, 2018

1st PQC Standardization conference



The only candidate based on LPN problem
Suitable for low-power devices even RFID

1st Candidate Submitted to NIST PQC

□ $A \xleftarrow{\$} Z_q^{m \times n}, x \xleftarrow{\$} Z_q^n, e \leftarrow \chi_q^m$, compute $y = Ax + e$

$$\begin{array}{c}
 A \\
 \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \\ a_{41} & a_{42} & a_{43} & a_{44} & a_{45} \\ a_{51} & a_{52} & a_{53} & a_{54} & a_{55} \\ a_{61} & a_{62} & a_{63} & a_{64} & a_{65} \\ a_{71} & a_{72} & a_{73} & a_{74} & a_{75} \\ a_{81} & a_{82} & a_{83} & a_{84} & a_{85} \end{bmatrix} \cdot \begin{array}{c} x \\ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \end{array} + \begin{array}{c} e \\ \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \\ e_8 \end{bmatrix} \end{array} = \begin{array}{c} y \\ \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \\ y_7 \\ y_8 \end{bmatrix} \end{array} \pmod{q}
 \end{array}$$

LPN: $q=2, \chi_q^{u,m} \sim \text{Bernoulli}$
Noise rate: u

LWE: $q \geq \text{poly}(n),$
 $\chi_q^m \sim \text{discrete Gaussian}$

□ Search LPN/LWE: find out x given $A, y = Ax + e$

LPN is the simplest version of the
hard learning problem family

1st Candidate Submitted to NIST PQC

Worst-case hardness [Brakerski et al., ePrint/2018/279]

- LPN (for certain params) at least as hard as a worst-case decoding problem

Average-case hardness

1. Constant-noise (μ independent of n)
 - BKW [Blum, Kalai, Wasserman, JACM 2003]: time $2^{O(n/\log n)}$
2. Low-noise $\mu \leq 1/\sqrt{n}$
 - Time $\text{poly}(n) \cdot e^{\mu n}$ with unbound many samples

Quantum hardness [Esser et al., Crypto 2017]

- Speedup (based on Grover search) $\leq$ quadratic over classic algorithms

Hardness of LPN

1st Candidate Submitted to NIST PQC

Adjusted	n	μn	C-Sec	Q-Sec	PK size	CT size	Err Prob.
Light I	$32767 \approx 2^{15}$	40	90	45	4KB	5.5KB	2^{-148}
Moderate I	$32767 \approx 2^{15}$	62	128	64	4KB	8KB	2^{-102}
Moderate II	2^{16}	90	192	96	8KB	13KB	2^{-110}

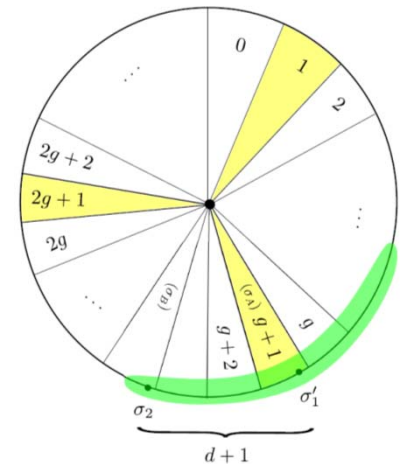
Main obstacle: public-key and ciphertext size

2nd Candidate Submitted to NIST PQC

KCL
(Key Consensus from Lattice)

Yunlei Zhao
Fudan University, Shanghai, China
(on behalf of the KCL team)

April 11, 2018



Optimal Key Consensus in
Presence of Noise.

2nd Candidate Submitted to NIST PQC

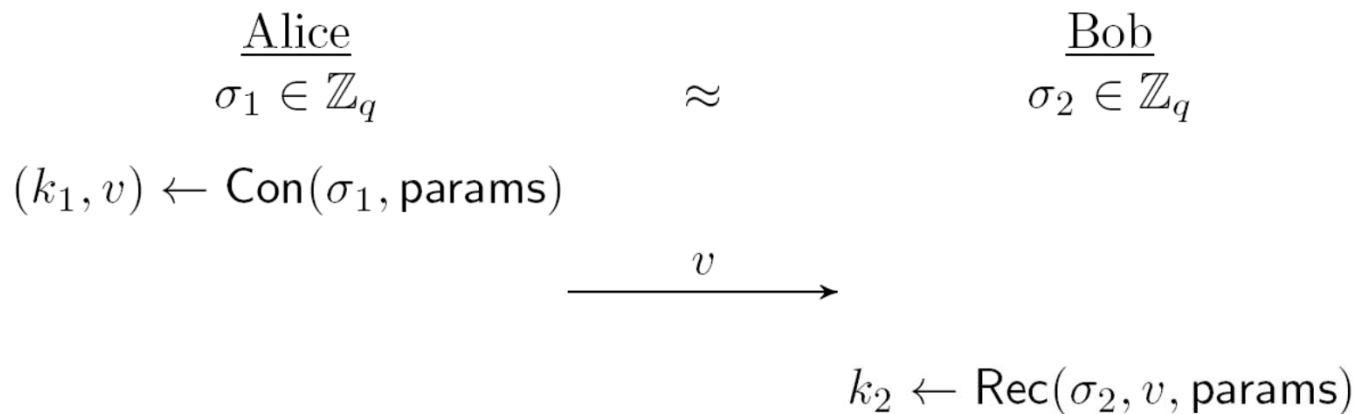


Figure: Depiction of KC

Completeness: $k_1 = k_2$, whenever $|\sigma_1 - \sigma_2|_q < d$.

Security: k_1 is independent of v , and is distributed uniformly at random over $\mathbb{Z}_m$, whenever $\sigma_1 \leftarrow \mathbb{Z}_q$.

Constraint: $2md \leq q \left(1 - \frac{1}{g}\right)$.

General Framework for PKE, KE

2nd Candidate Submitted to NIST PQC

	$ K $	bw.(B)	err.	pq-sec
OKCN-RLWE-SEC-1	765	3136	$2^{-68.4}$	250
OKCN-RLWE-SEC-2	765	3392	2^{-61}	258
NewHope	256	3872	2^{-61}	255
AKCN-RLWE-SEC-1	765	3264	$2^{-68.4}$	250
AKCN-RLWE-SEC-2	765	3520	2^{-61}	258
AKCN-RLWE-E8	512	3360	$2^{-63.3}$	262
NewHope-Simple	256	4000	2^{-61}	255

Table: Comparison between OKCN/AKCN-RLWE and NewHope. The actual NewHope proposal uses some smaller parameters, but with lowered security level.

KCL vs NewHope

3rd Candidate Submitted to NIST PQC

LAC

Lattice-based Cryptosystems

Principal Submitter: Xianhui Lu¹

Contact: luxianhui@iie.ac.cn, 0086-15010131069

Organizations: 1: Data Assurance and Communication Security Research Center,
Institute of Information Engineering, Chinese Academy of Sciences

2: OnBoard Security Inc.

Postal Address: No. 89A, Minzhuang Road, Haidian District, Beijing 100093, China

Auxiliary Submitters: Yamin Liu¹, Dingding Jia¹, Haiyang Xue¹, Jingnan He¹, Zhenfei Zhang²

Inventors: Xianhui Lu¹, Yamin Liu¹, Dingding Jia¹, Haiyang Xue¹, Jingnan He¹, Zhenfei Zhang²

The only byte-level modulus and bit-level noise
Ring-LWE based scheme

3rd Candidate Submitted to NIST PQC

NewHope: $n=1024, \sigma=\sqrt{8}, q=12289$

Kyber: $n=256*3, \sigma=2, q=7681$

LAC: $n=512, \sigma=1 / \sqrt{2}, q=251$

3rd Candidate Submitted to NIST PQC



`_mm256_maddubs_epi16`



$$c_1 = a_1 b_1 + a_2 b_2$$

||



AVX2 30 times speed up:
150 microseconds to 5 microseconds

3rd Candidate Submitted to NIST PQC

Scheme	Size (in Bytes)			AVX2 Cycles			Security
	sk	pk	ct	gen	enc	dec	
NewHope512	1888	928	1120	68,080	109,836	114,176	101
Kyber512	1632	736	800	55,160	75,680	74,428	102
LAC-128v2	1056	544	736	53,062	72,382	92,203	133
Kyber768	2400	1088	1152	85,472	112,660	108,904	161
LAC-192v2	2080	1056	1248	108,900	171,042	244,043	259
NewHope1024	3680	1824	2208	129,670	210,092	220,864	233
Kyber1024	3168	1440	1504	121,056	157,964	154,952	218
LAC-256v2	2080	1056	1472	97,464	174,512	323,207	290

1st Cryptanalysis of NIST PQC Candidate

From: YUYANG <y-y13@mails.tsinghua.edu.cn>
Sent: Thursday, April 19, 2018 9:05 PM
To: pqc-comments
Cc: pqc-forum@list.nist.gov
Subject: [pqc-forum] OFFICIAL COMMENT: DRS

Dear all,

Leo and I have updated our preprint on an attack against the DRS signature scheme:

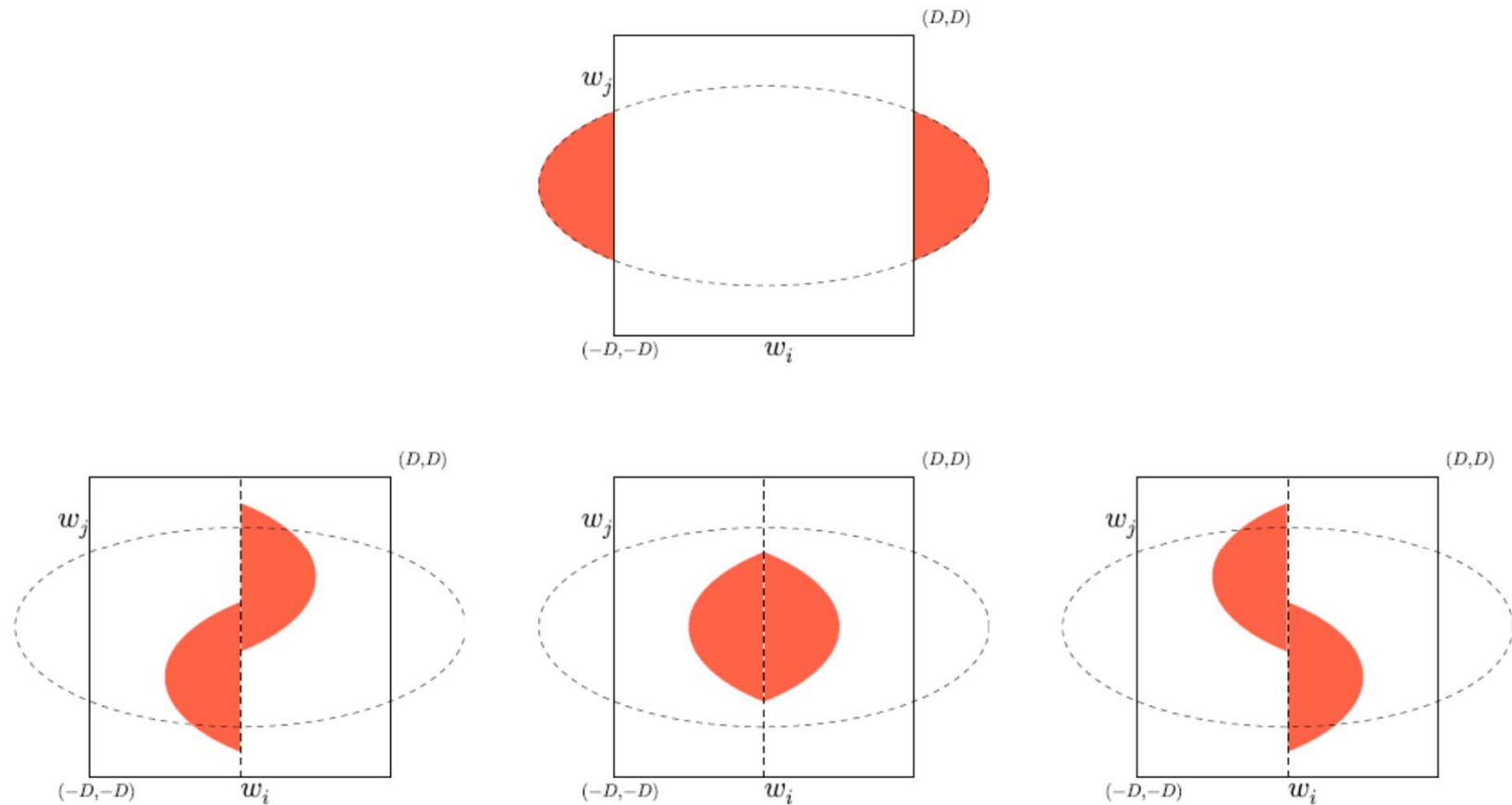
<https://eprint.iacr.org/2018/294.pdf>.

A link of open source scripts is provided in the updated version.

Best regards,
-- Leo Ducas and Yang Yu

Break DRS Scheme

1st Cryptanalysis of NIST PQC Candidate



statistical attack with deep learning

2rd Cryptanalysis of NIST PQC Candidate

Kerman, Sara J. (Fed)

From: Yanbin Pan <panyanbin@amss.ac.cn>
Sent: Wednesday, December 27, 2017 5:31 AM
To: pqc-comments
Cc: pqc-forum
Subject: OFFICIAL COMMENT: HK17
Attachments: HK17.pdf

Dear Dr. Chen, Moody, and Liu,

We group find that the key exchange scheme **HK17 is not secure**. Any passive adversary can recover the shared key very efficiently.

The key observation is that any octonion (or quaternion) satisfies a quadratic equation, so for any octonion o_A , and any polynomial $g(x)$, we can find a, b such that $g(o_A) = a o_A + b$.

For HK17, since $r_A = f(o_A)^m o_B f(o_A)^n$, we can write $r_A = (a o_A + b) o_B (c o_A + d)$ where a, b, c, d are unknowns. By solving a system of linear equations over $\mathbb{Z}_p$, we can find a solution a, b, c, d . Next we can prove that $(a o_A + b) r_B (c o_A + d)$ equals the shared key.

See the attachment for more details.

Best regards,
Yanbin

Break HK17 Scheme

2rd Cryptanalysis of NIST PQC Candidate

- ▶ Bernstein and Lange pointed out that the shared key can be recovered with $O(p)$ arithmetic operations.
- ▶ We independently propose an attack to show that just **constant** arithmetic operations are enough to recover the shared key for a passive adversary.

$$\mathbf{o}^2 + \alpha \mathbf{o} + \beta = 0. \quad g(\mathbf{o}) = a \cdot \mathbf{o} + b.$$

d	p	pre-quantum security (bits)	post-quantum security (bits)
16	251	64	32
32	65521	128	64
64	4294967291	256	128
128	184467440737	512	256

3rd Cryptanalysis of NIST PQC Candidate

From: Yanbin Pan <panyanbin@amss.ac.cn>
Sent: Saturday, January 06, 2018 9:47 AM
To: Tibouchi; Keita
Cc: pqc-comments; pqc-forum
Subject: Re: [pqc-forum] OFFICIAL COMMENT: Compact LWE

Dear all,

Thank Keita very much for sharing his code!

Our paper about the attack against NIST Compact-LWE has been posted by Cryptology ePrint Archive as <https://eprint.iacr.org/2018/020> .

Break Compact-LWE Scheme

3rd Cryptanalysis of NIST PQC Candidate

For $i \in \{1, 2, \dots, m\}$, generate Compact-LWE samples randomly as follows:

$$pk_i = \langle \mathbf{a}_i, \mathbf{s} \rangle + k^{-1} \cdot \overbrace{(sk \cdot u_i + r_i + e_i \cdot p)}^{\text{fairly smaller than } q} \bmod q$$
$$pk'_i = \langle \mathbf{a}_i, \mathbf{s}' \rangle + k'^{-1} \cdot \underbrace{(sk' \cdot u_i)}_{< 2^{54}} + \underbrace{(r'_i + e'_i \cdot p)}_{< 2^{56}} \bmod q (= 2^{64})$$

- **Public key:** $pk_i, pk'_i, \mathbf{a}_i$ and u_i .
- **Private key:** $\mathbf{s}, \mathbf{s}', k, k', p, sk, sk'$ along with ck, ck' satisfying $ck \cdot r_i + ck' \cdot r'_i = 0 \bmod p$

LWE with structured noise

Attend ISO/IEC SC27 WG2 SD8



ISO/IEC JTC 1/SC 27/WG 2 N 1811

ISO/IEC JTC 1/SC 27/WG 2

Cryptography and security mechanisms

Convenorship: JISC (Japan)

Document type: Standing Document

Title: WG 2 SD8 (Post-Quantum Cryptography) -- Part 3: Lattice-Based Mechanisms

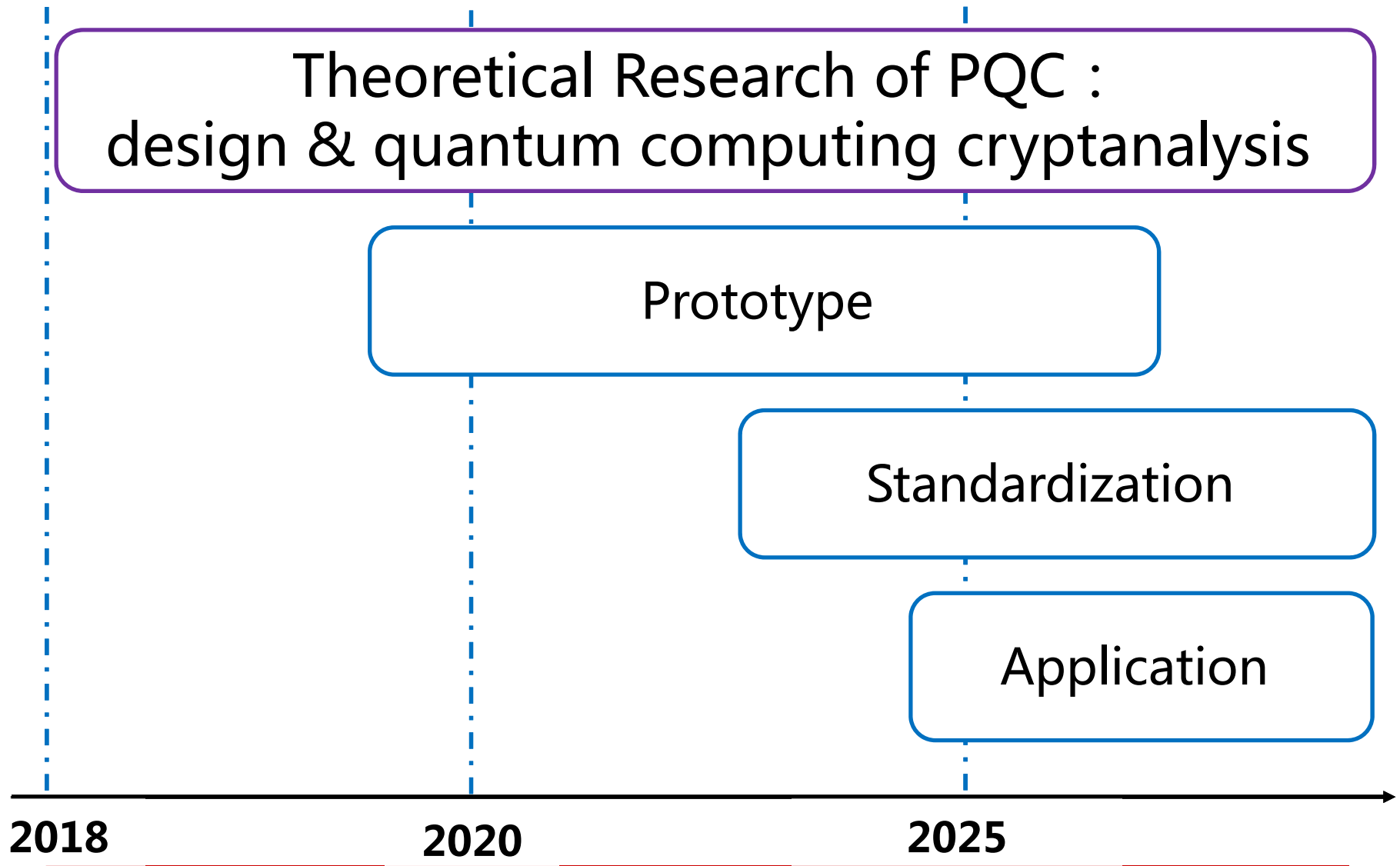
Status:

Date of document: 2018-09-28

Source: Editors (Xiannhui Lu, Le Trieu Phong, Zhenfei Zhang)

Attend the PQC project of ISO

Trends of PQC in China



Thanks!